

Lecture 10

Euclidean Inner Product

Norms, Orthogonal Vectors, Orthogonal Projections
Pythagoras Theorem, Least Square Approximation

Recall (from High School?)

Let $\mathbf{x}, \mathbf{y}$ belong to $\mathbb{R}^n$. What is the dot product of $\mathbf{x}$ and $\mathbf{y}$?

Euclidean Inner Product (Dot Product)

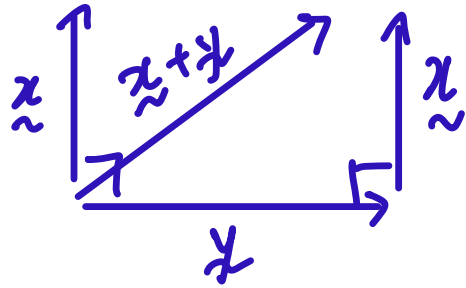
$$\langle \mathbf{x}, \mathbf{y} \rangle = \left\langle \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \right\rangle \triangleq x_1 y_1 + x_2 y_2 + \dots + x_n y_n \in \mathbb{R}$$

will get a number

① We will NOT use angle $\mathbf{x}$ and $\mathbf{y}$.

Recall (from High School?)

Let $\mathbf{x}, \mathbf{y}$ belong to $\mathbb{R}^n$. If $\mathbf{x}$ and $\mathbf{y}$ are orthogonal, what is $\|\mathbf{x} + \mathbf{y}\|^2$?



$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \Rightarrow \text{Pythagorean theorem}$$

Qn: How to define "orthogonal" without angles?

Lecture 10 – Euclidean Inner Product

- Norm, Orthogonality $\rightarrow$ define using $\langle, \rangle$
- Pythagoras Theorem $\rightarrow$ prove using new definitions
- Orthogonal Projection
- Least Squares Approximation (application of orthogonal projection)

Definition. Consider the vector space $\mathbb{R}^n$ and the Euclidean inner product of $\mathbf{x}$ and $\mathbf{y}$ is given by

$$\langle \mathbf{x}, \mathbf{y} \rangle \triangleq x_1 y_1 + \cdots + x_n y_n.$$

①

INPUTS: Two vectors

OUTPUT: Real number (in $\mathbb{R}$)

● : numbers

Proposition X. The Euclidean inner product obeys the following properties:

- (Positivity) $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$.
 - (Definiteness) $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$.
 - (Additivity in first slot) $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$ for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$.
 - (Homogeneity in first slot) $\langle \lambda \mathbf{x}, \mathbf{y} \rangle = \lambda \langle \mathbf{x}, \mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$.
 - (Symmetry) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
- } *diag*

Definition. Consider $\mathbb{R}^n$ and the Euclidean inner product of x and y is given by

$$\langle x, y \rangle \triangleq x_1 y_1 + \cdots + x_n y_n.$$

Proposition X (1/5). The Euclidean inner product obeys the following property:

- **(Positivity)** $\langle x, x \rangle \geq 0$ for all $x \in \mathbb{R}^n$.

(Positivity) For $x \in \mathbb{R}^n$,

$$\left\langle \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right\rangle = x_1^2 + x_2^2 + \cdots + x_n^2 \geq 0$$



$\therefore$ The Euclidean inner is positive. because x_i 's are real numbers

① Does NOT mean $\langle x, y \rangle \geq 0$ (i.e. $\langle x, y \rangle$ can be negative if x, y are different)

Definition. Consider $\mathbb{R}^n$ and the Euclidean inner product of $\mathbf{x}$ and $\mathbf{y}$ is given by

$$\langle \mathbf{x}, \mathbf{y} \rangle \triangleq x_1 y_1 + \cdots + x_n y_n.$$

Proposition X (2/5). The Euclidean inner product obeys the following property:

- (Definiteness) $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

$$(\Leftarrow) \mathbf{x} = \mathbf{0} \Rightarrow \langle \mathbf{x}, \mathbf{x} \rangle = x_1^2 + \cdots + x_n^2 = 0^2 + \cdots + 0^2 = 0$$

$$(\Rightarrow) \langle \mathbf{x}, \mathbf{x} \rangle = 0 \Rightarrow x_1^2 + \cdots + x_n^2 = 0$$

$$\Rightarrow x_1^2 = x_2^2 = \cdots = x_n^2 = 0 \Rightarrow x_1 = \cdots = x_n = 0$$

$$\Rightarrow \mathbf{x} = \mathbf{0}$$

[(!) $\langle \mathbf{x}, \mathbf{y} \rangle = 0$ does NOT imply
both $\mathbf{x}$ and $\mathbf{y}$ are $\mathbf{0}$]

Definition. Consider the vector space $\mathbb{R}^n$ and the Euclidean inner product of $\mathbf{x}$ and $\mathbf{y}$ is given by

$$\langle \mathbf{x}, \mathbf{y} \rangle \triangleq x_1 y_1 + \cdots + x_n y_n.$$

The Euclidean norm of $\mathbf{x}$ is given by $\|\mathbf{x}\| \triangleq \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$

$$\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle = x_1^2 + \dots + x_n^2$$

$$\therefore \|\mathbf{x}\| = \sqrt{x_1^2 + \dots + x_n^2}$$

= usual definition of length of vector

Proposition. Consider the Euclidean norm.

Suppose $x \in \mathbb{R}^n$.

(a) $\|x\| = 0$ if and only if $x = \mathbf{0}$.

(b) $\|\lambda x\| = |\lambda| \|x\|$ for $\lambda \in \mathbb{R}$.

Definition. $\|x\| \triangleq \sqrt{\langle x, x \rangle}$.

"(a) means that the only vector of length zero is $\mathbf{0}$ ".

$$\|x\| = 0$$

$$\Rightarrow \|x\|^2 = 0$$

$$\Rightarrow \langle x, x \rangle = 0$$

$$\Rightarrow x = \mathbf{0}$$

Definiteness of Euclidean
Inner Product

Proposition. Consider the Euclidean norm.

Suppose $x \in \mathbb{R}^n$.

(a) $\|x\| = 0$ if and only if $x = \mathbf{0}$.

(b) $\|\lambda x\| = |\lambda| \|x\|$ for $\lambda \in \mathbb{R}$.

Definition. $\|x\| \triangleq \sqrt{\langle x, x \rangle}$.

"(b) means that if we scale x by λ ,
then its length scale by $|\lambda|$ "

$$\|\lambda x\|^2 = \langle \lambda x, \lambda x \rangle$$

Homogeneous in first slot $= \lambda \langle x, \lambda x \rangle$

Symmetry $= \lambda \langle \lambda x, x \rangle$

$$= \lambda^2 \|x\|^2$$

Take square root,

$$\|\lambda x\| = |\lambda| \cdot \|x\|$$

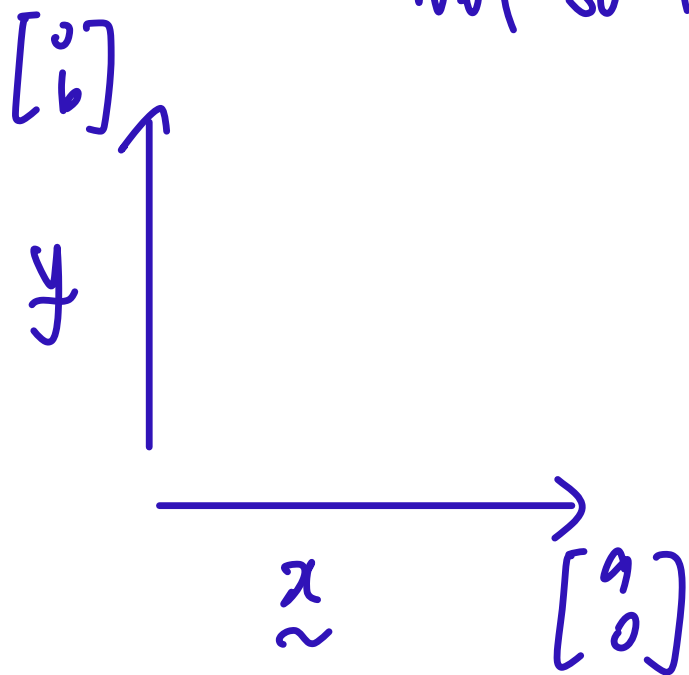
λ might be negative,
must put absolute sign

Definition. Consider $\mathbb{R}^n$ and the Euclidean inner product of $\mathbf{x}$ and $\mathbf{y}$ is given by

$$\langle \mathbf{x}, \mathbf{y} \rangle \triangleq x_1 y_1 + \cdots + x_n y_n.$$

Definition. In $\mathbb{R}^n$, two vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$.

Not-so-interesting example



$$n=2. \quad \langle \tilde{x}, \tilde{y} \rangle = \langle \begin{bmatrix} a \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ b \end{bmatrix} \rangle = 0$$

$\therefore \tilde{x}, \tilde{y}$ are orthogonal

Definition. Consider $\mathbb{R}^n$ and the Euclidean inner product of x and y is given by
$$\langle x, y \rangle \triangleq x_1 y_1 + \cdots + x_n y_n.$$

Definition. In $\mathbb{R}^n$, two vectors x and y are orthogonal if $\langle x, y \rangle = 0$.

$$x = \begin{bmatrix} 3 \\ -4 \end{bmatrix}, \quad y = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\langle x, y \rangle = 12 - 12 = 0$$

$\therefore x, y$ is orthogonal

$$x + y = \begin{bmatrix} 7 \\ -1 \end{bmatrix}$$

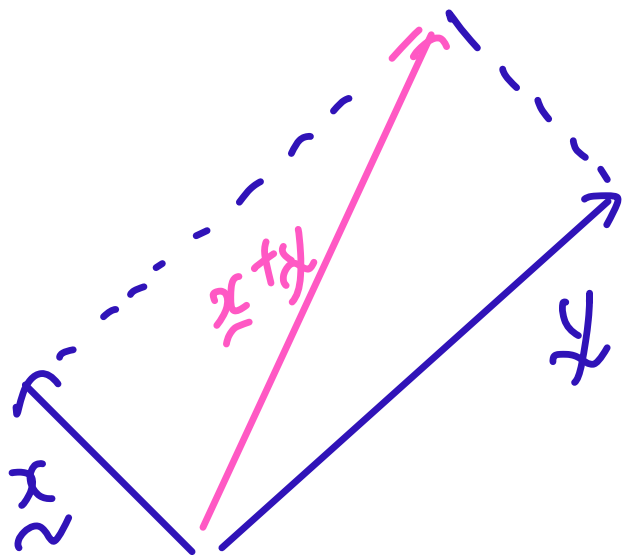
$$\|x + y\|^2 = 50$$

$$\|x\|^2 = 25 \quad \|y\|^2 = 25$$

$$\therefore \|x\|^2 + \|y\|^2 = \|x + y\|^2$$

$\hookrightarrow$ Pythagorean Theorem.

$\star \| \cdot \| \rightarrow$ for vector
 $| \cdot | \rightarrow$ for number



Pythagoras Theorem. If x and y are orthogonal, then

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2.$$

Definition. Consider $\mathbb{R}^n$ and let $\langle x, y \rangle$ denote the Euclidean inner product.

- Two vectors x and y are orthogonal if $\langle x, y \rangle = 0$.
- $\|x\| = \sqrt{\langle x, x \rangle}$.

Proof $\|x+y\|^2 = \langle x+y, x+y \rangle$

$$= \langle x, x+y \rangle + \langle y, x+y \rangle$$

Additive in first slot.

$$= \langle x+y, x \rangle + \langle x+y, y \rangle$$

$$= \langle x, x \rangle + \langle y, x \rangle + \langle x, y \rangle + \langle y, y \rangle$$

$$\|x\|^2$$

$$= \|x\|^2 + \|y\|^2$$

$\Rightarrow 0$
as they are orthogonal

$$\|y\|^2$$

Observe!

- All results (including Pythagoras Theorem)
are proved using Proposition X only

Specifically, in all proofs, we did NOT use the fact
 $\langle x, y \rangle = x_1 y_1 + \dots + x_n y_n$

Proposition X. The Euclidean inner product obeys the following properties:

- **(Positivity)** $\langle x, x \rangle \geq 0$ for all $x \in \mathbb{R}^n$.
- **(Definiteness)** $\langle x, x \rangle = 0$ if and only if $x = 0$.
- **(Additivity in first slot)** $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$ for all $x, y, z \in \mathbb{R}^n$.
- **(Homogeneity in first slot)** $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$ for all $x, y \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$.
- **(Symmetry)** $\langle x, y \rangle = \langle y, x \rangle$ for all $x, y \in \mathbb{R}^n$.

- We see the implications
NEXT WEEK.

Properties of Euclidean Inner Product

Proposition. Consider the Euclidean inner product.

Fix $\mathbf{v} \in \mathbb{R}^n$ and consider the map $\mathbf{T}$ that maps from $\mathbb{R}^n$ to $\mathbb{R}$ such that $\mathbf{T}(\mathbf{u}) = \langle \mathbf{u}, \mathbf{v} \rangle$. Then $\mathbf{T}$ is a linear map.

Furthermore, we have $\langle \mathbf{v}, \mathbf{0} \rangle = \langle \mathbf{0}, \mathbf{v} \rangle = 0$ for $\mathbf{v} \in \mathbb{R}^n$.

→ proof in colab

Prove these two results using Prop. X

Proposition. Consider the Euclidean inner product.

Then the following holds.

- **(Additivity in second slot)** For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, we have that

$$\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle.$$

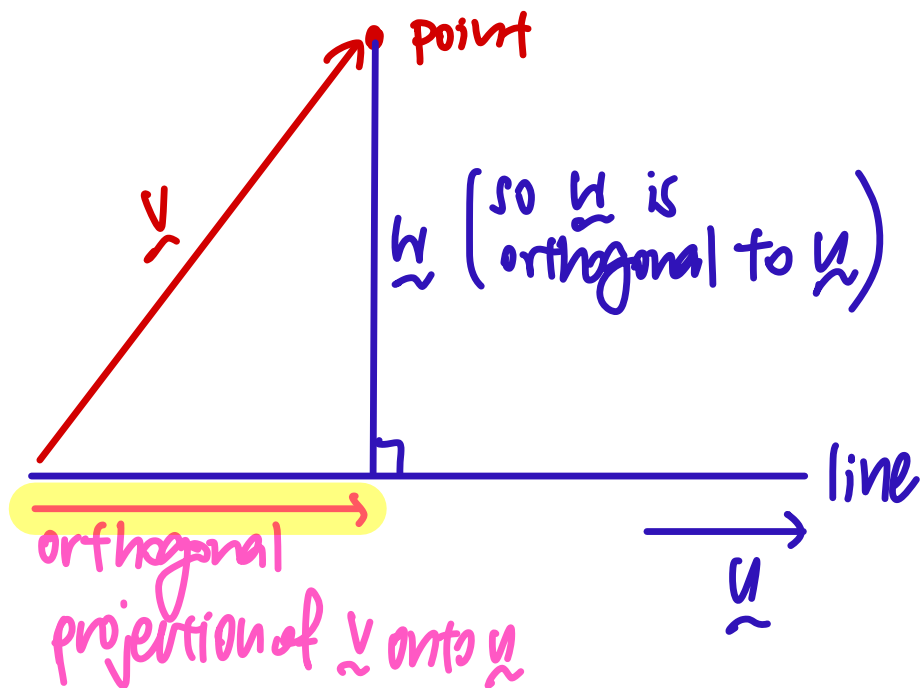
- **(Homogeneity in second slot)** For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and for $\lambda \in \mathbb{R}$,

$$\langle \mathbf{u}, \lambda \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle.$$

} Int 9, Q1

Orthogonal Projection

Orthogonal Projections



$$= \text{Proj}_{\underline{u}}(\underline{v})$$

$$\underline{v} = \underline{u} + \underline{w} \quad (*)$$

Take inner product of (*) with $\underline{u}$

$$\begin{aligned} \langle \underline{v}, \underline{u} \rangle &= \langle \underline{u} + \underline{w}, \underline{u} \rangle \\ &= \underline{u} \langle \underline{u}, \underline{u} \rangle + \langle \underline{w}, \underline{u} \rangle \end{aligned}$$

$\rightsquigarrow 0$

$$\Rightarrow \underline{u} = \frac{\langle \underline{v}, \underline{u} \rangle}{\langle \underline{u}, \underline{u} \rangle}$$

$$\therefore \text{Proj}_{\underline{u}}(\underline{v}) = \frac{\langle \underline{v}, \underline{u} \rangle}{\langle \underline{u}, \underline{u} \rangle} \underline{u}$$

Formula only involves inner products.
 $\hookrightarrow$ no angles!

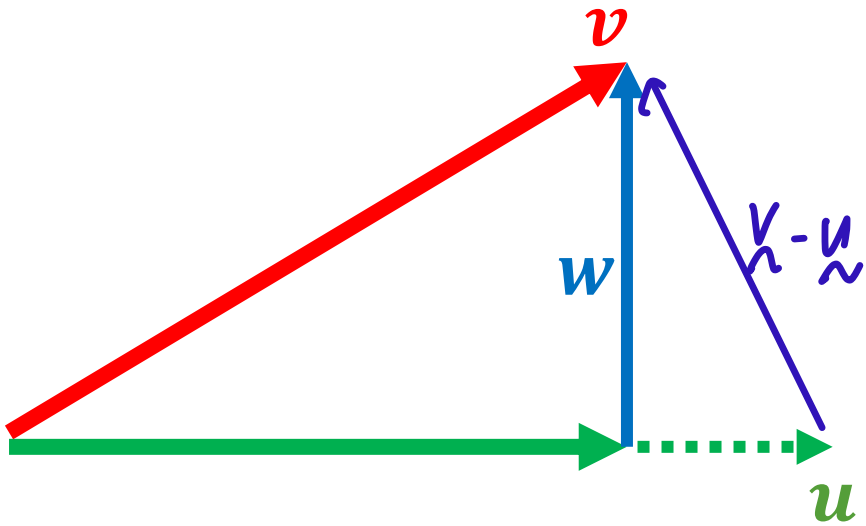
Orthogonal Projections

JC: we learn that the closest point on $\mathcal{U}$ to $\mathbf{x}$ is

$$\text{Proj}_{\mathcal{U}}(\mathbf{x})$$

$$\text{i.e. } \|\mathbf{w}\| = \|\mathbf{x} - \text{Proj}_{\mathcal{U}}(\mathbf{x})\| \leq \|\mathbf{x} - \mathbf{u}\|$$

let's prove this using new notion of orthogonality.



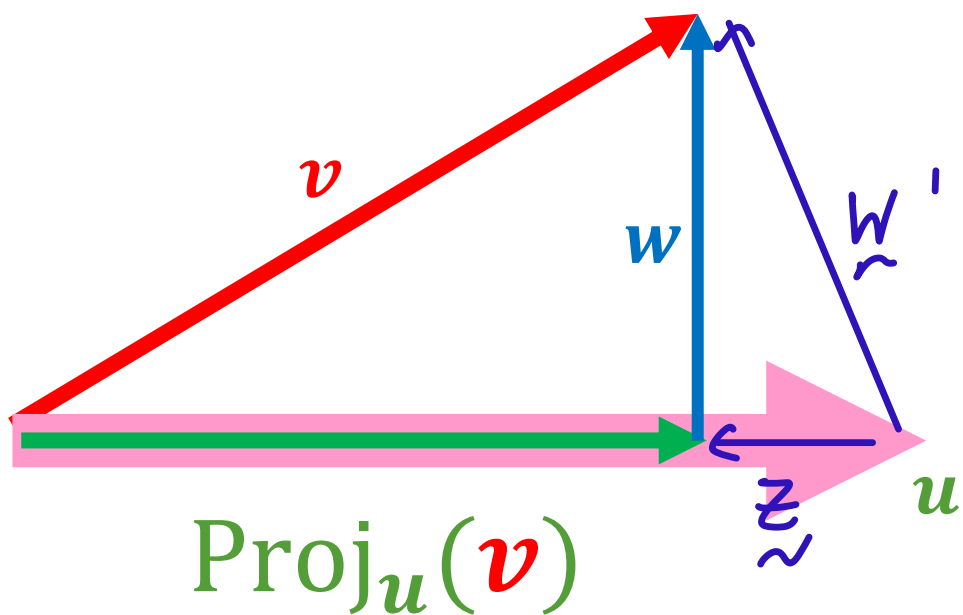
$$\text{Proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \mathbf{u}$$

Proposition. Let $\mathcal{V}$ be an inner product vector space.

Let u, v belong to $\mathcal{V}$.

Then

$$\|w\| = \|v - \text{Proj}_u(v)\| \leq \|v - u\| = \|w'\|$$



WTS: $\|w\| \leq \|w'\|$

– z is orthogonal to w

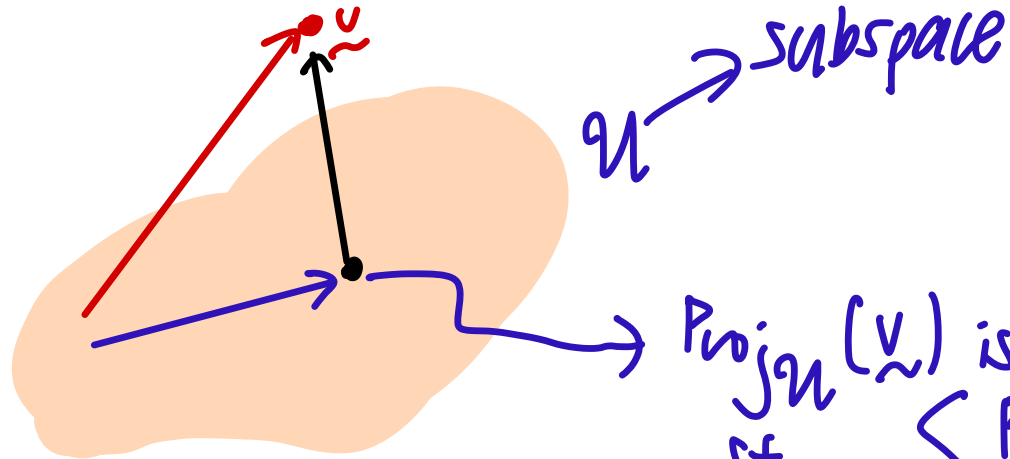
– $z + w = w'$

By Pythagoras Thm,

$$\|w'\|^2 = \|z\|^2 + \|w\|^2$$

→ Because $\|z\|^2 \geq 0$,
we have
 $\|w'\|^2 \geq \|w\|^2$

Orthogonal Projection onto a Subspace



$\text{Proj}_U(\underline{v})$ is defined to be the unique vector in U
st. $\langle \text{Proj}_U(\underline{v}) - \underline{v}, \underline{u} \rangle = 0$ for all $\underline{u}$ in U

Proposition. Consider $\mathbb{R}^n$. Let U be a subspace of $\mathbb{R}^n$.

Then

$$\|v - \text{Proj}_U(v)\| \leq \|v - u\| \text{ for all } u \in U.$$

Moreover, $\text{Proj}_U(v)$ is THE vector in U that is “closest” to v .

* Why is vector unique?
→ see text for proof.

Proof omitted.

Formula for orthogonal projection

Theorem. Consider the Euclidean inner product.

Consider a subspace $\mathcal{U}$ and let $\mathcal{A} = [\mathbf{u}_1, \dots, \mathbf{u}_k]$ be an orthogonal basis for $\mathcal{U}$.

Let $\mathbf{v} \in \mathcal{V}$. Then

$$\text{Proj}_{\mathcal{U}}(\mathbf{v}) = \frac{\langle \mathbf{v}, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 + \frac{\langle \mathbf{v}, \mathbf{u}_2 \rangle}{\langle \mathbf{u}_2, \mathbf{u}_2 \rangle} \mathbf{u}_2 + \dots + \frac{\langle \mathbf{v}, \mathbf{u}_k \rangle}{\langle \mathbf{u}_k, \mathbf{u}_k \rangle} \mathbf{u}_k.$$

- A basis $[\mathbf{u}_1, \dots, \mathbf{u}_k]$ is orthogonal if $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ if $i \neq j$

→ Discuss more next week.

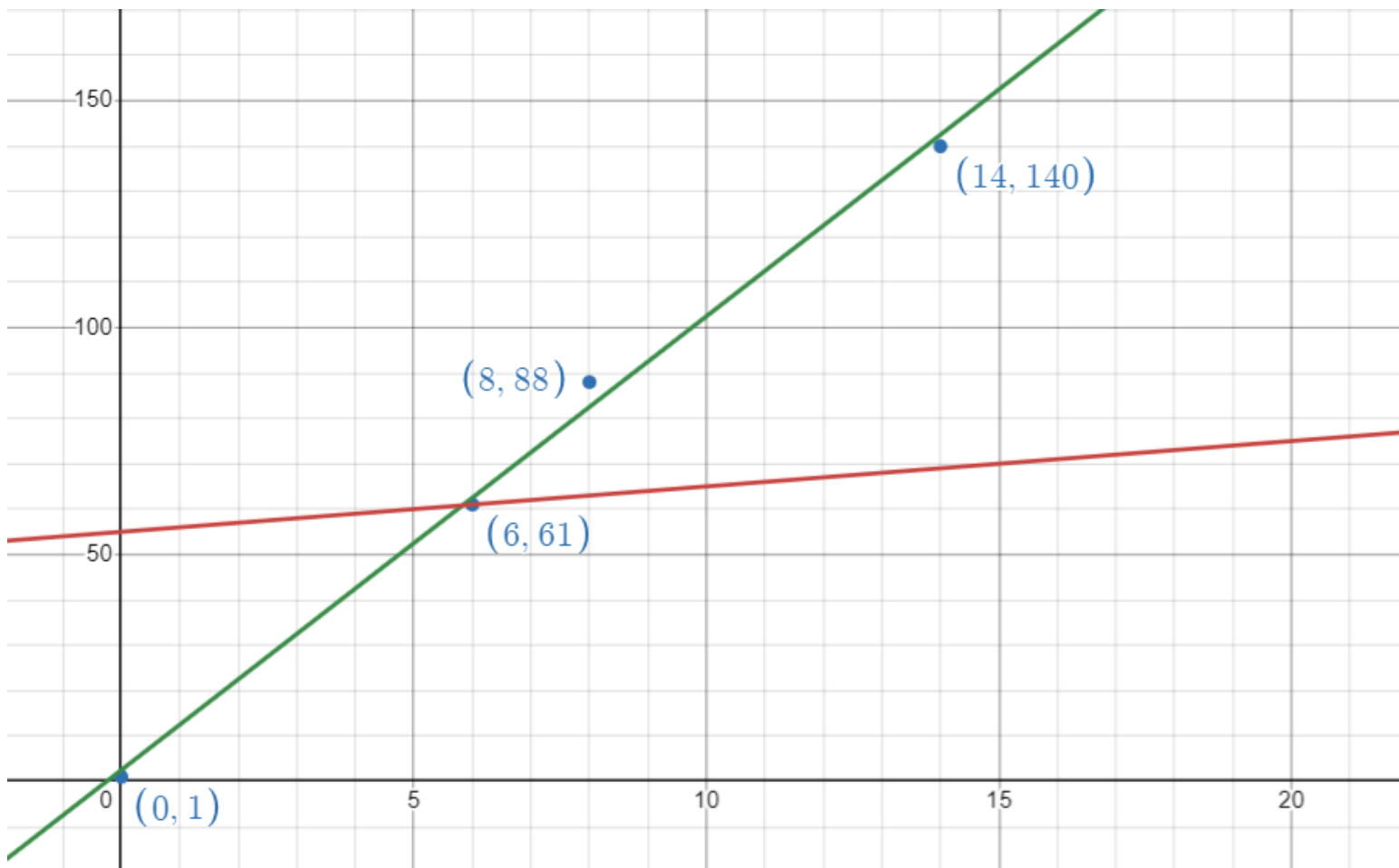
→ Proof of theorem in colab

An Application of Orthogonal Projection: Least Squares Approximation

Least Square Approximation

--The Problem

wooclap



Consider the task of “fitting a straight line” through four blue data points.

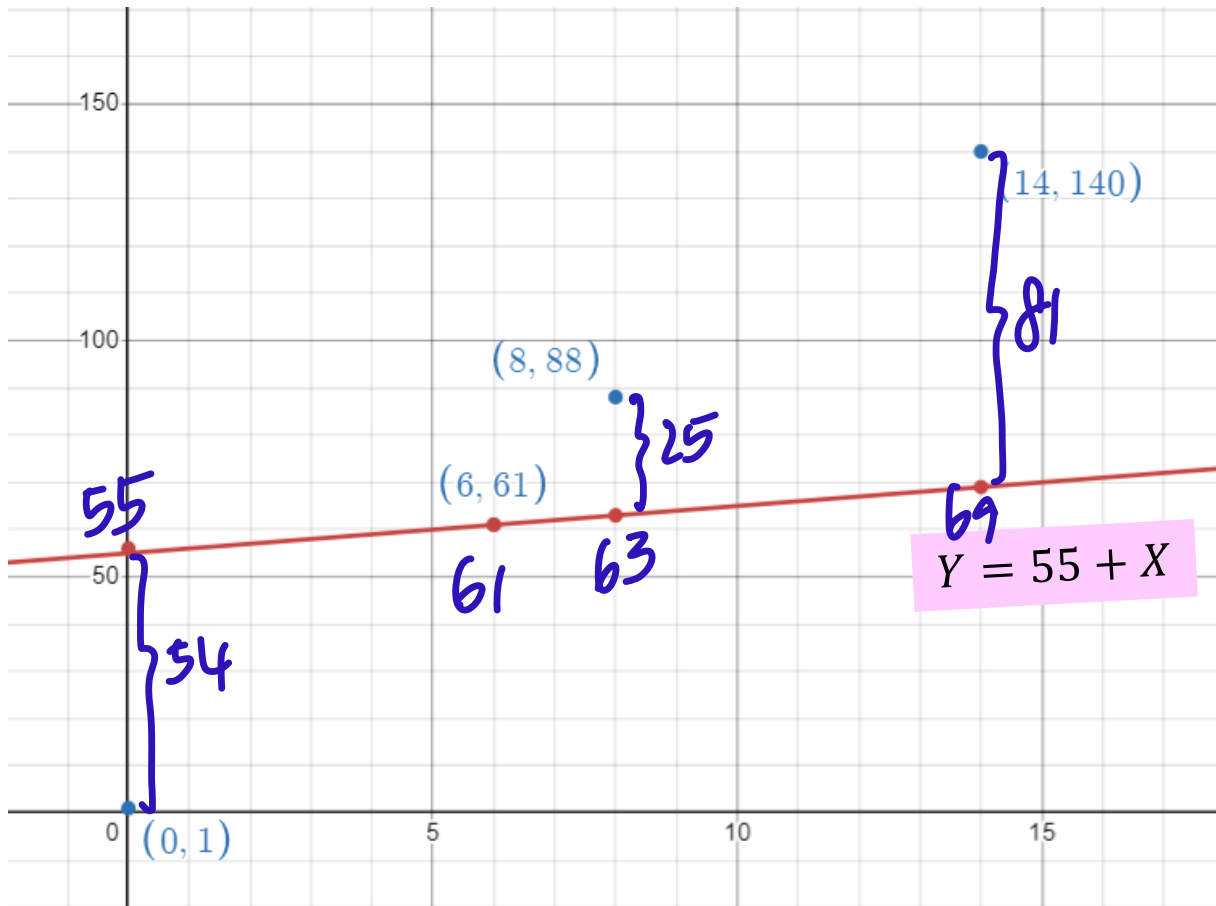
Which line is better?

↳ green line

→ Let's “quantify” the goodness of fit.

Least Square Approximation

--How Good is an Approximation?



Suppose we use the line $Y = X + 55$.

Let the sum squared error (SSE)

$$= (55 - 1)^2 + (61 - 61)^2 + (63 - 88)^2 + (69 - 140)^2$$

● : guess values (on the line)

● : actual data

Least Square Approximation

--How to Compute SSE?

wooclap

Consider the following vectors.

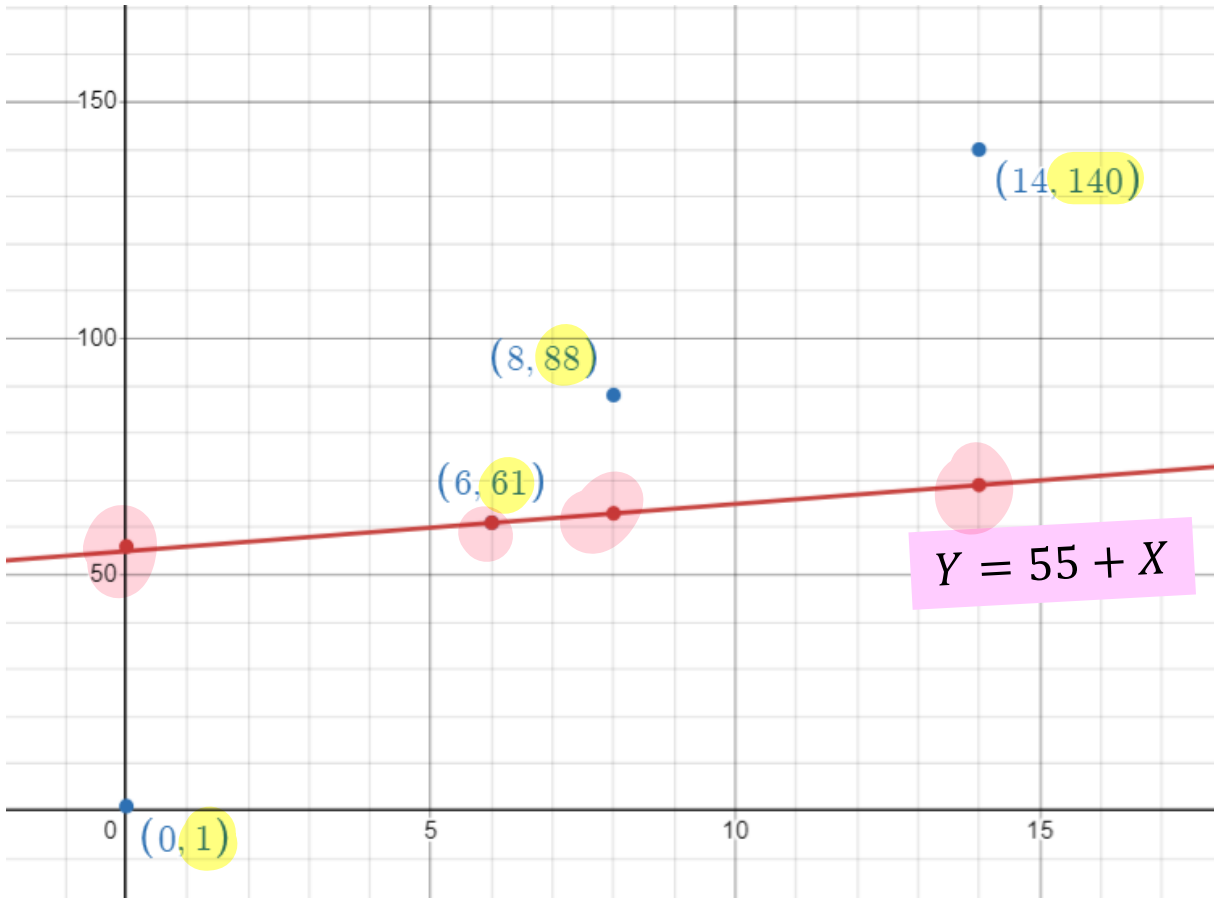
$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{a}_2 = \begin{bmatrix} 0 \\ 6 \\ 8 \\ 14 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 61 \\ 88 \\ 140 \end{bmatrix}$$

Consider the vector

$$\mathbf{u} = 55\mathbf{a}_1 + \mathbf{a}_2 = \begin{bmatrix} 55 \\ 61 \\ 63 \\ 69 \end{bmatrix}$$

How to compute the SSE?

$$SSE = \|\mathbf{u} - \mathbf{b}\|^2$$

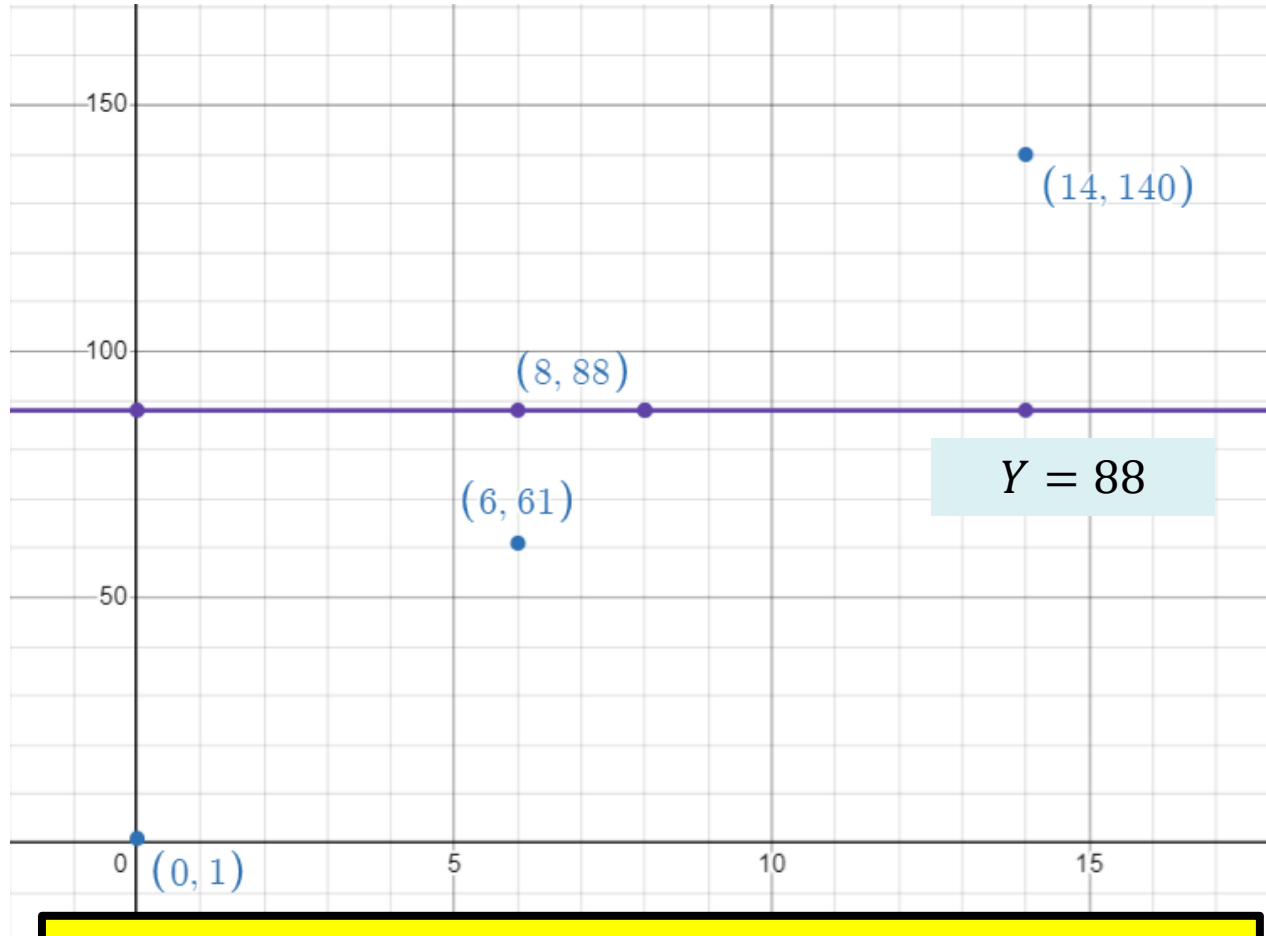


SSE

$$= (55 - 1)^2 + (61 - 61)^2 + (63 - 88)^2 + (69 - 140)^2$$

Least Square Approximation

--How to Compute SSE?



SSE

$$= (88 - 1)^2 + (88 - 61)^2 + (88 - 88)^2 + (88 - 140)^2$$

Consider the following vectors.

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{a}_2 = \begin{bmatrix} 0 \\ 6 \\ 8 \\ 14 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 61 \\ 63 \\ 140 \end{bmatrix}$$

Consider the vector

$$\begin{bmatrix} 88 \\ 88 \\ 88 \\ 88 \end{bmatrix} = \mathbf{u} = \lambda_1 \mathbf{a}_1 + \lambda_2 \mathbf{a}_2 = \begin{bmatrix} \lambda_1 + 0 \\ \lambda_1 + 6\lambda_2 \\ \lambda_1 + 8\lambda_2 \\ \lambda_1 + 14\lambda_2 \end{bmatrix}$$

What should λ_1 and λ_2 be s.t. $\lambda_1 = 88, \lambda_2 = 0$

$$\text{SSE} = \|\mathbf{u} - \mathbf{b}\|^2?$$

more generally, If $Y = MX + C$,
Choose $\lambda_1 = C, \lambda_2 = M$.

Least Square Approximation

--Equivalent Problem is...

wooclap

Consider the following vectors.

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{a}_2 = \begin{bmatrix} 0 \\ 6 \\ 8 \\ 14 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 61 \\ 63 \\ 140 \end{bmatrix}$$

For each λ_1 and λ_2 ,

$$\text{set } \mathbf{u} = \lambda_1 \mathbf{a}_1 + \lambda_2 \mathbf{a}_2$$

Then our task is to find λ_1 and λ_2 that minimizes $\|\mathbf{u} - \mathbf{b}\|^2$.

What is an equivalent statement?

Find $\mathbf{u} \in \text{span}(\mathbf{a}_1, \mathbf{a}_2)$
s.t. $\|\mathbf{u} - \mathbf{b}\|^2$ is minimised.

Theorem (Linear Regression).

Consider the points $\{(\alpha_1, \beta_1), (\alpha_2, \beta_2), \dots, (\alpha_m, \beta_m)\}$.

Set $\mathbf{a}_1 = (1, 1, \dots, 1)^T$, $\mathbf{a}_2 = (\alpha_1, \alpha_2, \dots, \alpha_m)^T$ and $\mathcal{U} = \text{span}(\{\mathbf{a}_1, \mathbf{a}_2\})$.

Then the line $Y = \lambda_1 + \lambda_2 X$ minimizes SSE if and only if

$$\text{Proj}_{\mathcal{U}}(\mathbf{b}) = \lambda_1 \mathbf{a}_1 + \lambda_2 \mathbf{a}_2. \quad \text{where } \underline{\mathbf{b}} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}$$

→ Equiv
 $\hat{\mathbf{A}} = \begin{bmatrix} \uparrow & \uparrow \\ \hat{\mathbf{a}}_1 & \hat{\mathbf{a}}_2 \\ \downarrow & \downarrow \end{bmatrix}$

Then a vector in $\mathcal{U} = \text{span}(\mathbf{a}_1, \mathbf{a}_2)$
is of the form $\underline{\mathbf{u}} = \hat{\mathbf{A}} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \hat{\mathbf{A}} \underline{\mathbf{x}}$

$$\min \|\underline{\mathbf{u}} - \underline{\mathbf{b}}\|^2 \text{ s.t. } \underline{\mathbf{u}} \in \mathcal{U} \text{ is equiv. to } \min \|\hat{\mathbf{A}} \underline{\mathbf{x}} - \underline{\mathbf{b}}\|^2 \text{ s.t. } \underline{\mathbf{x}} \in \mathbb{R}^n$$

↓
also least squares approx

Theorem (Least Squares Approximation).

Consider an $m \times n$ -matrix A and $\mathbf{b} \in \mathbb{R}^m$.

Suppose that $\mathbf{x} \in \mathbb{R}^n$ minimizes $\|A\mathbf{x} - \mathbf{b}\|^2$.

Then $A^T A \mathbf{x} = A^T \mathbf{b}$. ← solving this equation

Let us do linear regression on

$(0,1), (6,61), (8,88), (14,140)$

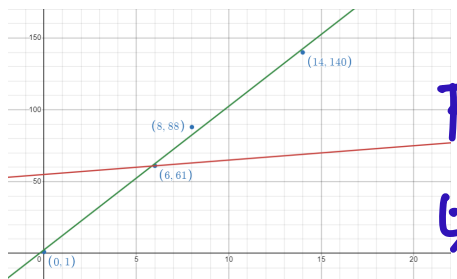
Consider the following vectors.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 6 \\ 1 & 8 \\ 1 & 14 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 61 \\ 88 \\ 140 \end{bmatrix}.$$

Linear Regression: choose $\hat{A}$

Fit a parabola/quadratic:

↳ Tut 9, Q8



$$\overset{2 \times 4}{\hat{A}^T} \overset{4 \times 2}{\hat{A}} = \begin{bmatrix} 4 & 28 \\ 28 & 296 \end{bmatrix}$$

$$\overset{2 \times 4}{\hat{A}^T} \overset{4 \times 1}{\mathbf{b}} = \begin{bmatrix} 290 \\ 3030 \end{bmatrix}$$

$$\text{Solving } \begin{bmatrix} 4 & 28 \\ 28 & 296 \end{bmatrix} \begin{bmatrix} \pi_1 \\ \pi_2 \end{bmatrix} = \begin{bmatrix} 290 \\ 3030 \end{bmatrix}$$

$$\Rightarrow \pi_1 = 2.5, \pi_2 = 10$$

Theorem (Least Squares Approximation).

Consider an $m \times n$ -matrix A and $\mathbf{b} \in \mathbb{R}^m$.

Suppose that $\mathbf{x} \in \mathbb{R}^n$ minimizes $\|A\mathbf{x} - \mathbf{b}\|^2$.

Then $A^T A \mathbf{x} = A^T \mathbf{b}$.

Let's solve $\hat{A}^T \hat{A} \underline{x} = \hat{A}^T \mathbf{b}$ again

$$\hat{R}^T \hat{Q}^T \hat{Q} \hat{R} \underline{x} = \hat{R}^T \hat{Q}^T \mathbf{b}$$

$$\hat{R}^T \hat{R} \underline{x} = \hat{R}^T \begin{bmatrix} 145 \\ 100 \end{bmatrix} \Rightarrow \hat{R} \underline{x} = \begin{bmatrix} 145 \\ 100 \end{bmatrix} \Rightarrow \text{Solve: } \begin{bmatrix} 2 & 14 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 145 \\ 100 \end{bmatrix}$$

$\hat{R}^T$ is invertible

$$\Rightarrow x_2 = 10$$

① This whole procedure does NOT involve $\lambda_1 = 2.5$ inverting matrices.
TA;PI: Generally computing matrix inverses should be avoided.

Let us do linear regression on

$(0,1), (6,61), (8,88), (14,140)$

Consider the following vectors.

$$\hat{A} = \hat{Q} \hat{R} \quad A = \begin{bmatrix} 1 & 0 \\ 1 & 6 \\ 1 & 8 \\ 1 & 14 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.7 \\ 0.5 & -0.1 \\ 0.5 & 0.1 \\ 0.5 & 0.7 \end{bmatrix} \begin{bmatrix} 2 & 14 \\ 0 & 10 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 \\ 61 \\ 88 \\ 140 \end{bmatrix}.$$

$$\hat{Q} \xrightarrow{\text{special}} \hat{Q}^T \hat{Q} = \hat{I}$$

upper triangular matrix

Theorem (QR factorization).

Consider a matrix A whose columns are linearly independent.
Then we can write

$$A = QR$$

such that

- $Q^T Q = I$; *→ Here we say that $\hat{Q}$ is orthogonal.*
- R is upper triangular.

*Next week: we find $\hat{Q}$ and $\hat{R}$
using Gram-Schmidt*